

# Homework 3

Caleb Neely

## Set up

### Problem 1

#### a) An Appropriate Test

To determine the strength of the relationship between ocean temperature (°C) and giant kelp frond elongation rate (cm day<sup>-1</sup>), appropriate tests would be a **simple linear regression** and a **Pearson correlation**. Simple linear regression evaluates whether ocean temperature can predict giant kelp elongation rate and estimates how much elongation rate changes with temperature, while Pearson correlation measures the strength and direction of the linear relationship between the two continuous variables.

#### b) Visualization

```
ggplot(temp_kelp, aes(x = temp_c, y = kelp_elong)) +  
  
  geom_point(  
    color = "darkgreen",  
    size = 1  
  ) +  
  
  geom_smooth(  
    method = "lm",  
    se = TRUE,  
    color = "steelblue"  
  ) +  
  
  labs(  
    x = "Ocean temperature (°C)",
```

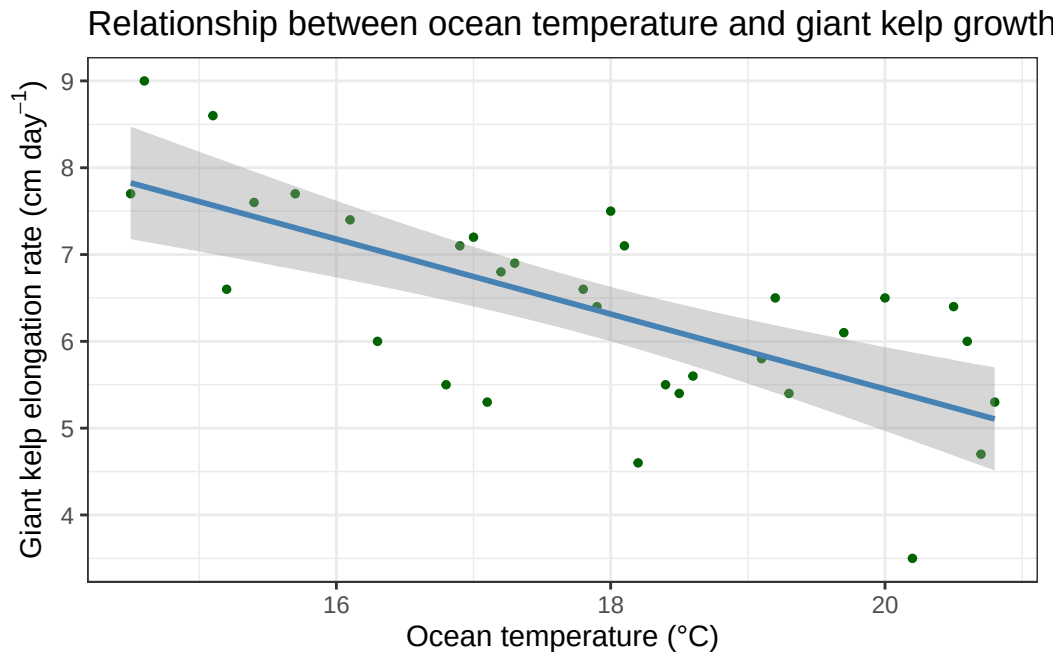
```

y = expression("Giant kelp elongation rate (cm day-1)"),
title = "Relationship between ocean temperature and giant kelp growth"
) +

theme_bw()

```

`geom\_smooth()` using formula = 'y ~ x'



### c) Check Assumptions And Run Test

#### Assumption Checks

```

# visualize relationship to check linearity and outliers

ggplot(temp_kelp, aes(x = temp_c, y = kelp_elong)) +

  geom_point(
    color = "darkgreen",
    size = 1
  ) +

```

```

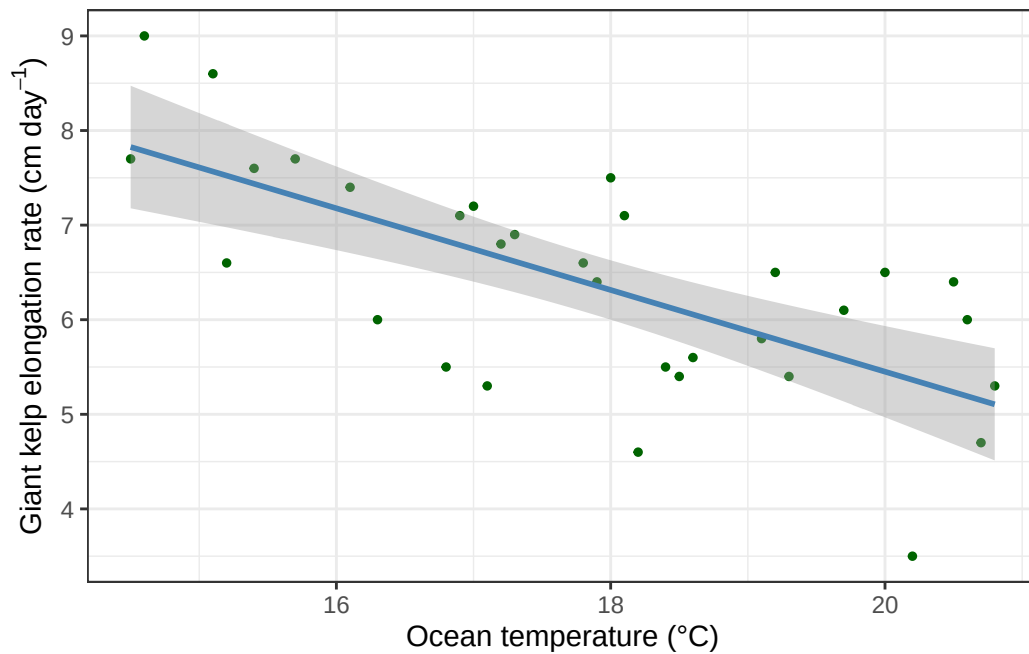
geom_smooth(
  method = "lm",
  se = TRUE,
  color = "steelblue"
) +

labs(
  x = "Ocean temperature (°C)",
  y = expression("Giant kelp elongation rate (cm day"^-1*)")
) +

theme_bw()

```

`geom\_smooth()` using formula = 'y ~ x'



```

# check normality of variables

temp_kelp %>%
  shapiro_test(temp_c, kelp_elong)

```

```

# A tibble: 2 x 3

```

|   | variable   | statistic | p     |
|---|------------|-----------|-------|
|   | <chr>      | <dbl>     | <dbl> |
| 1 | kelp_elong | 0.988     | 0.966 |
| 2 | temp_c     | 0.963     | 0.322 |

```
# create linear model

kelp_lm <- lm(kelp_elong ~ temp_c, data = temp_kelp)

# check residual normality

shapiro_test(residuals(kelp_lm))
```

```
# A tibble: 1 x 3
  variable      statistic p.value
  <chr>         <dbl>   <dbl>
1 residuals(kelp_lm) 0.951   0.159
```

I checked the assumptions of linearity, normality, and outliers for the linear regression model and Pearson correlation test. I checked linearity and potential outliers using a scatterplot with a fitted regression line, and I checked normality using Shapiro-Wilk tests from **rstatix** on both variables and the regression residuals. The assumptions appeared reasonably met because the relationship looked linear and the residuals were approximately normally distributed.

## Run Test

```
# run linear regression model

kelp_lm <- lm(kelp_elong ~ temp_c, data = temp_kelp)

summary(kelp_lm)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = temp_kelp)
```

Residuals:

|  | Min     | 1Q      | Median | 3Q     | Max    |
|--|---------|---------|--------|--------|--------|
|  | -1.8643 | -0.5017 | 0.1790 | 0.5659 | 1.2177 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 14.08645 | 1.49219    | 9.440   | 1.72e-10 *** |
| temp_c      | -0.43179 | 0.08321    | -5.189  | 1.37e-05 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

#### d) Results communication

To evaluate the relationship between ocean temperature and giant kelp frond elongation rate, I used a **simple linear regression** because I wanted to determine whether ocean temperature could predict changes in giant kelp growth rate.

The linear regression showed a significant negative relationship between ocean temperature and giant kelp elongation rate ( $p = 0.0000137$ ). This indicates that giant kelp frond elongation rate decreased as ocean temperature increased, suggesting that warmer ocean conditions may negatively affect giant kelp growth.

#### e) Test implications

I would communicate to the team that ocean temperature appears to have a significant negative relationship with giant kelp frond elongation rate, meaning kelp growth decreased as temperature increased. This suggests that warming ocean conditions associated with climate change could negatively affect giant kelp productivity and overall kelp forest health, which may have broader impacts on the marine ecosystems that depend on kelp forests.

#### f) Run Other Test

```
# Pearson correlation test

temp_kelp %>%
  cor_test(
    temp_c,
    kelp_elong,
    method = "pearson"
  )
```

```
# A tibble: 1 x 8
  var1    var2      cor statistic      p conf.low conf.high method
  <chr>  <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  <chr>
1 temp_c kelp_elong -0.69      -5.19 0.0000137   -0.836   -0.446 Pearson
```

The simple linear regression and Pearson correlation would lead to the same decision about the null hypothesis because both evaluate the strength of the linear relationship between ocean temperature and giant kelp elongation rate. However, linear regression estimates how much elongation rate changes with temperature, while Pearson correlation specifically measures the strength and direction of the linear association between the two continuous variables. In this case, both tests indicate that giant kelp elongation rate decreases as ocean temperature increases.